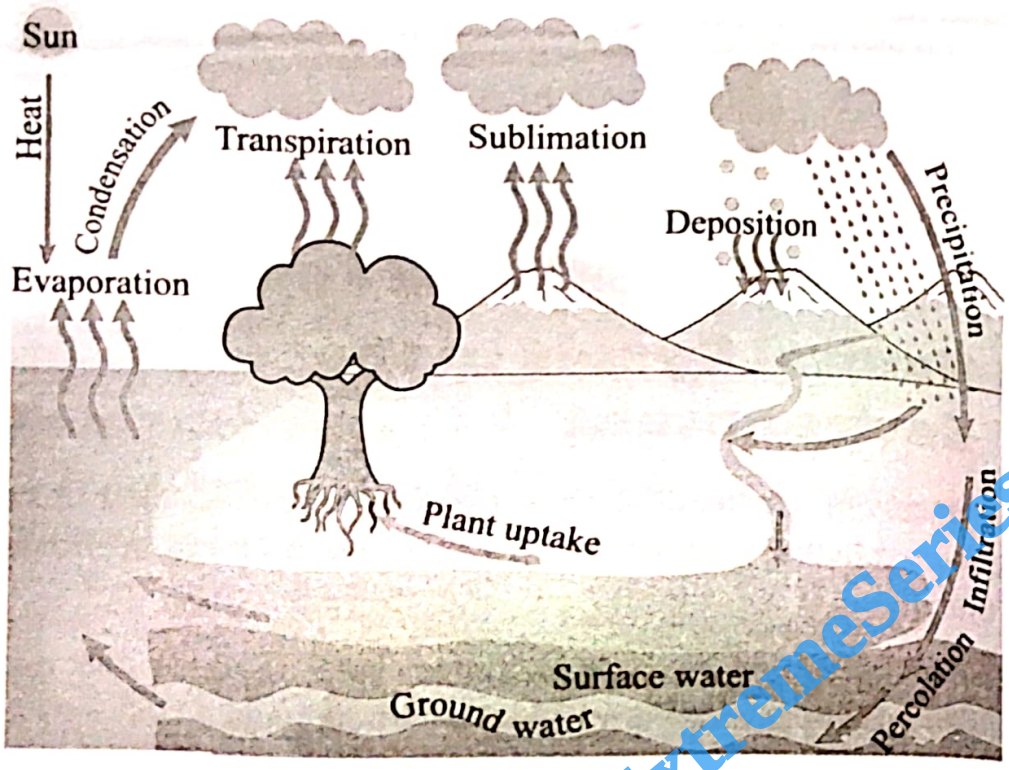


Fig 11 . 3 The water Cycle

2. The Oxygen Cycle

- Oxygen is essential for life of plants and animals
- The oxygen cycle involves photosynthesis by plants and respiration by all living organisms. It maintains a constant level of oxygen found in the environment.

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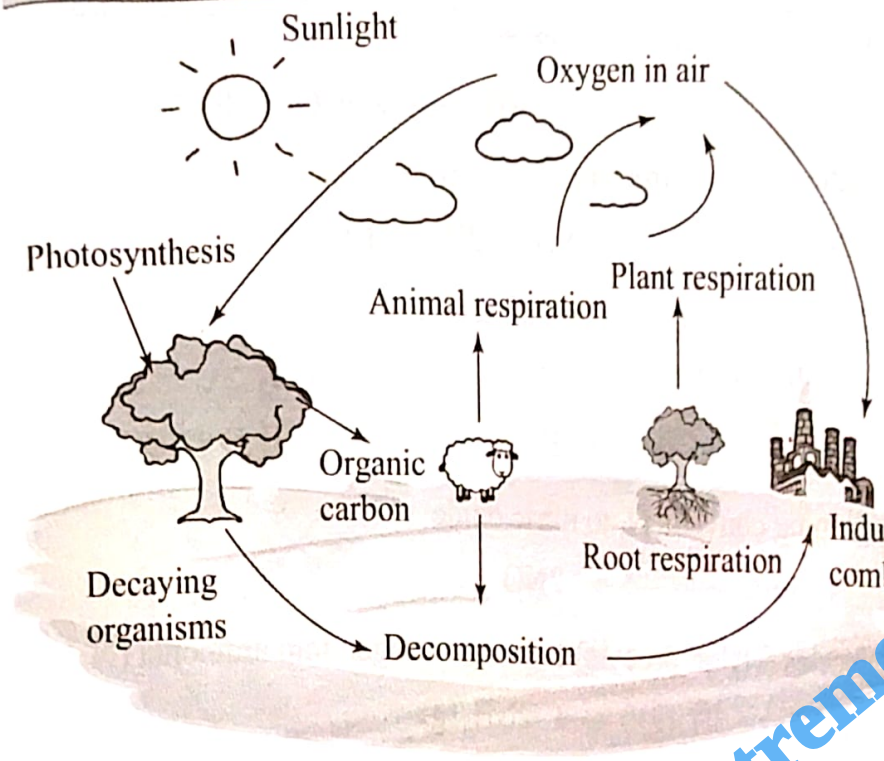


Fig 11 . 4 The Oxygen Cycle

The Oxygen Cycle is useful in:

- Photosynthesis:- $6CO_2 + 6H_2O \xrightarrow{\text{Sun light}} C_6H_{12}O_6 + 6O_2$
- Respiration:- $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$
- Energy production: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{heat}$
- Weathering of minerals: $4FeO + O_2 \longrightarrow 2Fe_2O_3$
- Ozone formation: $O + O_2 \longrightarrow O_3$

3. The Nitrogen Cycle

Nitrogen gas can be industrially obtained by **fraction distillation of liquid air**

Nitrogen is an essential element for living organisms.

Nitrogen is important to make protein. This can be done by **plants.** Animals obtain the nitrogen they need for protein production by feeding plants or other animals.

Nitrogen fixation is the direct use of atmospheric nitrogen in the formation of nitrogen compounds.

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- Fixation by bacteria and algae

$$2N_2 + 3CH_2O + 3H_2O + 4H^+ \longrightarrow 3CO_2 + 4NH_4^+$$

- Nitrification by nitrosomas and nitrobacteria

$$2O_2 + 3NH_4^+ \longrightarrow NO_3^- + 2H^+ + H_2O$$

- Nitrate reduction by microbial action

$$2NO_3^- + CH_2O \longrightarrow 2NO_2^- + H_2O + CO_2$$

$$2NO_2^- + 3\,CH_2O + 4H^+ \longrightarrow 2NH_4^+ + H_2O + 3CO_2$$

- Lightning converts N_2 to nitric oxide

$$N_2 + O_2 \xrightarrow{\text{lightining}} 2NO$$

- Factories (Haber process) converts nitrogen into ammonia (NH_3)

$$2N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

- **Denitrification** returns nitrogen into the atmosphere.

$$2NO_3^- + 5CH_2O + 4H^+ \longrightarrow 2N_2 + 7H_2O + 5CO_2$$

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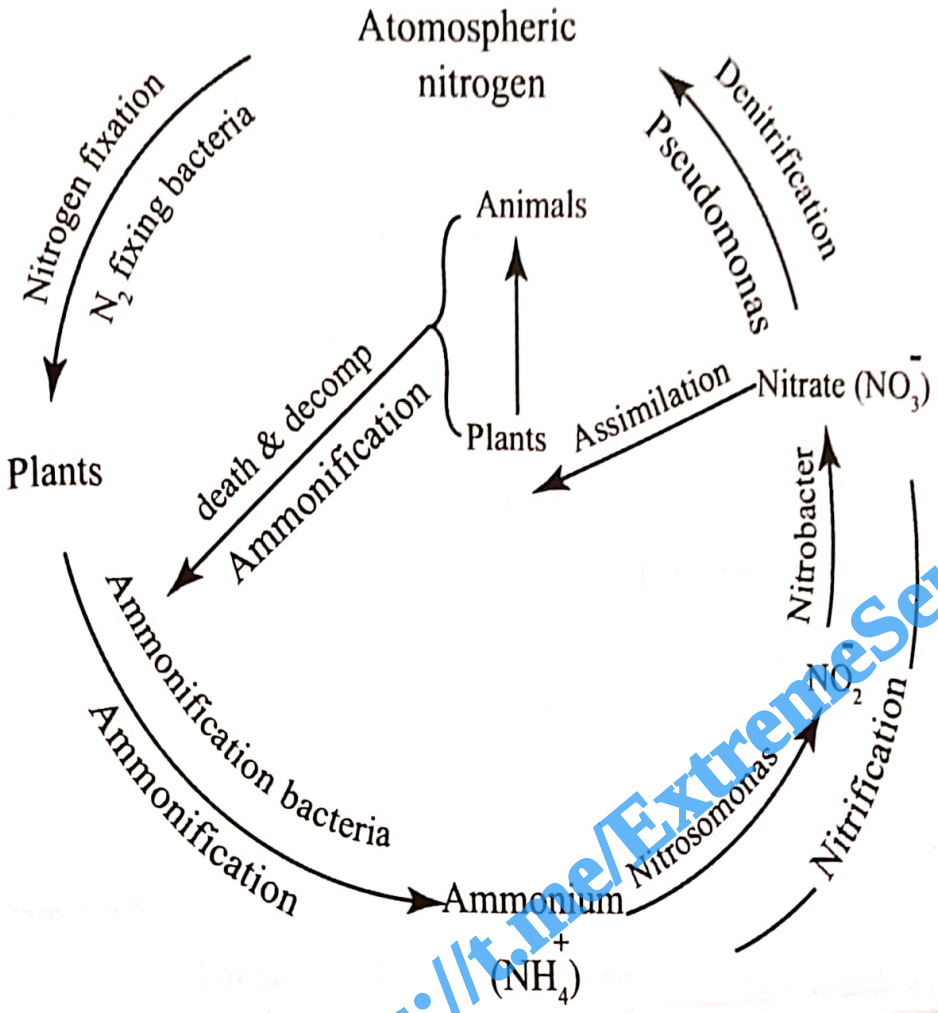


Fig 11 . 5 The nitrogen cycle

Illustrative Examples

14. All of following add water vapour to the atmosphere. **EXCEPT**

A. Evaporation
B. Respiration
C. Transpiration
D. Photosynthesis

Solution

Photosynthesis consumes water and releases oxygen to the atmosphere.

Answer: D

15. The driving force behind the water cycle is

A. Evaporation
B. Condensation
C. Combustion
D. Heat of the sun

Answer: D

16. Which of the following processes does not reduce the amount of

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oxygen from the atmosphere?

A. Respiration
C. Combustion
B. Photosynthesis
D. Weathering of mineral

Solution

Photosynthesis is a process that adds oxygen to the atmosphere.

Answer: B

17. Which of the following reactions in the oxygen cycle represents degradation of organic material?

A. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

B. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat

C. $4FeO + O_2 \rightarrow 2Fe_2O_3$

D. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Solution

B) Energy production

C) Weathering of minerals

D) Photosynthesis

Answer: A

18. Which of the following absorbs ultraviolet radiations?

A. O_2 B. N_2 C. CO_2 D. O_3

Answer: D

19. The conversion of nitrogen gas to nitrates by bacteria is called

A. Nitrogen fixation
C. Execretion
B. Nitrification
D. Dentrification

Solution

Nitrogen fixation is a biological or industrial process that converts atmospheric nitrogen to nitrate.

Answer: B

20. Write one artificial method that converts nitrogen into nitrogen compound.

Solution

Haber process: $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

21. The process of converting nitrogen compounds back into nitrogen is known as _____

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Answer: Denitrification

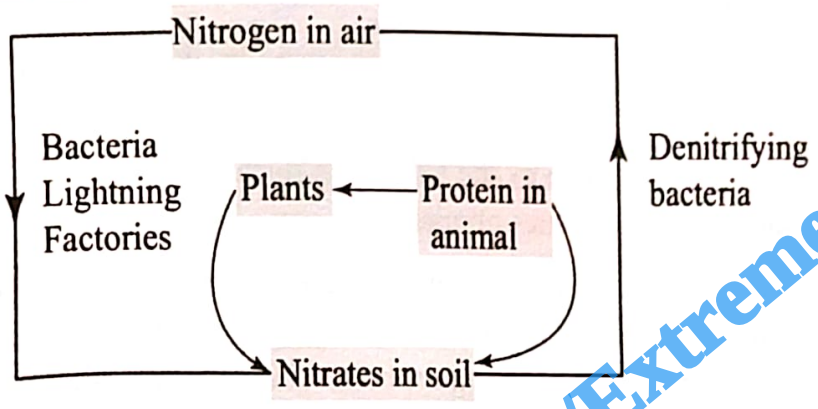
22. Write two natural processes that convert nitrogen into nitrogen compounds.

Solution

- Nitrifying bacteria
- Lightning

23. Describe the nitrogen cycle using diagram.

Solution



4. The Phosphorus Cycle

☑ Phosphorus is not found in the atomsphere

☑ Living cells use phosphorus to make RNA, DNA and ATP.

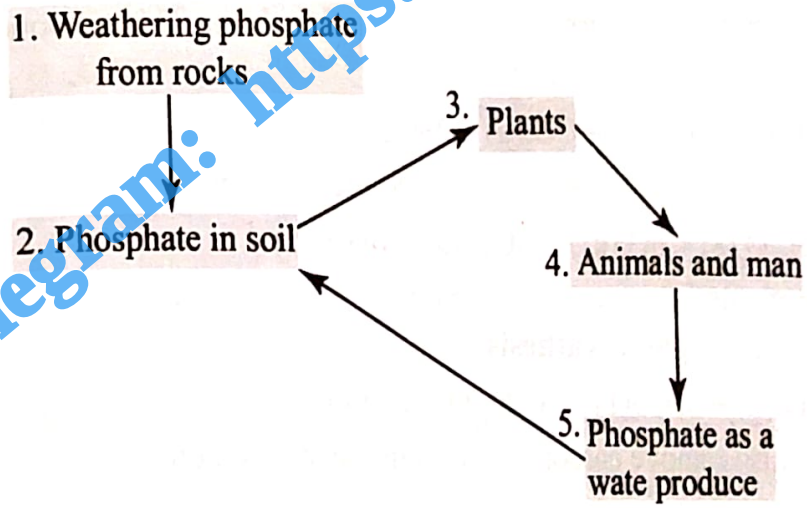


Fig 11 . 6 The Phosphorus Cycle

5. The Sulphur Cycle

☑ Sulphur is a yellow non - metallic element that is found uncombined in layers under the ground.

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☑ Industrially, it is manufactured by **Frasch process.**

☑ About 90% of the sulphur extracted is used to manufacture sulfuric acid. The rest is used to vulcanize rubber.

The sulphur Cycle can be Explained as Follows

☑ The sulphur is released by the weathering of rocks.

☑ Sulphur comes in contact with air and is converted into sulphates.

☑ Sulphates are taken up by plants and microbe and are converted into organic forms.

☑ The organic form of sulphur is then consumed by the animals through their food and thus sulphur moves in the food chain.

☑ When animals die, some of the sulphur is released by decomposition while some enter the tissues of microbes.

☑ There are several natural sources such as volcanic eruptions. evaporation of water and break down of organic matter in swamps, that release sulphur directly into atmosphere. This sulphur falls on earth with rainfall.

6. The Carbon Cycle

☑ The sea, the atmosphere, the rocks and living things contain carbon

☑ Most fuels produce carbondioxide when they burn. This carbondixide escapes in to the atmosphere.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(1)$$

☑ Respiration also adds carbondioxide to the atmosphere.

$$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(1)$$

☑ Green plants remove large amounts of carbondioxide from the atmosphere by **photosynthesis**

$$6CO_2(g) + 6H_2O(1) \rightarrow C_6H_{12}O_6(s) + 6O_2(g)$$

☑ Oceans also remove carbonioxide from the atmosphere.

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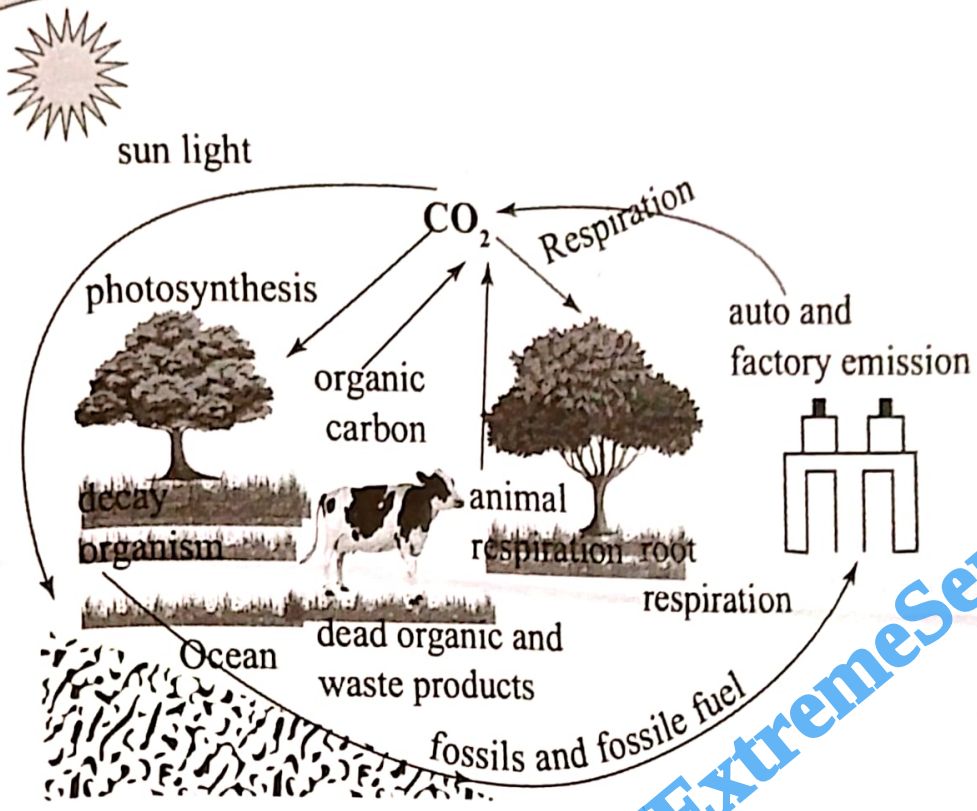


Fig 11 . 7 The Carbon Cycle

Illustrative Examples

24. In what way is the phosphorus cycle differs from carbon and nitrogen cycles?

A. Phosphors is more abundant that carbon and nitrogen.

B. Phosphorus is less important to biological system than carbon or nitrogen

C. Phosphorus, once used by an organism does not cycle back to the environment.

D. The reserviors for phosphorus exists in mineral rocks than in the atmosphere.

Solution

Unlike carbon and nitrogen cycles, the phosphorus cycle is slow process and phosphorus is not found in the atmosphere.

Answer: D

25. Which one of the following is needed for photosynthesis?

A. CO_2 B. N_2 C. O_2 D. P_4

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Solution

The raw materials for photosynthesis are CO_2 and H_2O

Answer: A

26. Which of the following substances is manufactured industrially by Frasch process?

A. Carbon B. Phosphorus C. Sulphur D. Nitrogen

Answer: C

27. Which of the following processes removes carbondioxide from the atmosphere?

A. Photosynthesis
B. Factories
C. Respiration
D. Vehicles.

Solution

Photosynthesis and ocean remove carbondioxide from the atmosphere.

Answer: A

28. The main processes that keep the amounts of carbondioxide in the atmosphere constant are _____ and _____

Solution

Photosynthesis and respiration

29. The solubility of phosphorus in water can be controlled by ____ and ___

Solution

Fe and Al (in acidic medium) and Ca in basic conditions.

30. Why is deforestation harmful to the environment?

Explanation

Plants remove carbondioxide from the atmosphere for photosynthesis. Removing plants increases the amount of carbondoxide in the atmosphere and causes global warming.

11.3 Environmental Pollution

- ☑ **Pollution** is the discharge of harmful waste materials to the environment.
- ☑ Materials that cause pollution are called **pollutants.**
- ☑ A substance that is introduced into the environment through human activity is called **contaminant**. It is considered as pollutant if it has significant effect on human health.

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☑ The medium that converts pollutants chemically is called **sink**.

☑ **Dissolved Oxygen (DO)** is oxygen dissolved in water which is important for aquatic life.

Table 11.2 Amount of DO and quality of water

Amount of DO in mg/L	Quality of water
4 - 8	Good quality water
Less than 4	Polluted water, not fit for human and animal consumption

☑ **Biological Oxygen Demand (BOD)** is the measure of the amount of dissolved oxygen in the biological process of metabolizing organic matter in water.

☑ It is determined experimentally by measuring Do at beginning and end of a 5 - day period in a sealed sample

☑ **Threshold Limit Value(TLV)** is a recommended occupational exposure limit for a chemical substance or physical agent.

1. Air Pollution

➥ It is mainly due to combustion reaction.

The Common Air Pollutants are:

i. Sulphurdioxide (SO_2) causes acid rain

$$SO_2 + O_2 \longrightarrow SO_3$$

$$SO_3 + H_2O\ (rain) \longrightarrow H_2SO_4(aq)$$

Acid rain

ii. Oxides of nitrogen - causes acid rain

$$N_2 + O_2 \xrightarrow{\text{Lightning}} 2NO$$

$$2NO + O_2 \longrightarrow 2NO_2$$

$$NO_2 + H_2O\ (rain) \longrightarrow 2HNO_3 + NO$$

Acid rain

iii. Oxides of carbon

Carbonmonoxide is formed due to an incomplete burning.

It is a poisonous gas that combines with the pigment in the cells of the blood that carry oxygen around the body.

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Carbodioxide is formed by complete combusion.

- ➥ It causes acid rain and increases the temperature of an environment.

2. **Hydrocarbons and ozone** are responsible for photochemical smog.
3. **Particulates** - irritate the lung and affect breathing
4. **Chlorofluorocarbons (CFC_s)** deplet the ozone layer.
5. **Heavy metals** - Lead is hazardous if it gets into our food or water supply or if we breathe it in. Even small amount of lead in the body can cause damage to the nervous system, kidneys and liver.
6. **Volatile organic compound (VOCs)** - converted into gas or vapor easily
7. **Poly chlorinated biphenyls (PCBs)** - are organic chlorine compounds

Methods of Reducing Air pollution

- ➥ Recycling and reusing materials and avoiding plastic bags.
- ➥ Plant trees
- ➥ Prevent wild fires
- ➥ Particulate filters for diesel vehicles.
- ➥ Removing of sulphurdioxide from gaseous fuels (flue - gas desulphurization) or use low sulphur
- ➥ Using public transports, carpooling or biking instead of driving.
- ➥ Avoiding burning leaves, trash, wood and crackers
- ➥ Turning off the lights and appliances when not in use.
- ➥ Use renewable energy to power house

2. Water Pollution

☑ Every day, hundreds of harmful substances find their way into our rivers oceans, lakes and seas. This causes **water pollution.**

The Water Pollutants Include

- ➥ Fertilizers
- ➥ Untreated sewage
- ➥ Detergents
- ➥ Oil
- ➥ Waste from factories
- ➥ Heat
- ➥ Lead

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Methods of Reducing Water Pollution

- ☑ Treating waste before dischanging in to water bodies.
- ☑ Industries should not discharge hot water that heats tanks and pipes in to the river.
- ☑ Recycling industrial and agricultural wastes.
- ☑ Using natural ferfilizers and biological method of controlling pests.

3. Land Pollution

- ☑ It is caused by the discharge of wastes on it.

Land pollutants include.

i. Dumping of non - biodegradable substance such as plastics
ii. Leaching of harmful chemicals and metals ions.
iii. Excessive use of synthetic fertilizers and chemicals
iv. Spillage of oil from leaking pipe lines.

Methods of Reducing Land Pollution

- ➥ Manufacture biodegradable plastics
- ➥ Recycling wastes
- ➥ Using natural fertilizers
- ➥ Converting potentially dangerous chemical wastes into harmless substances.
- ➥ Using the waste of one industry as a raw materials in another industry.

Table 11.3 Main Causes or Agents of Air, Water and Land Pollutions

Main causes of air pollution	Main causes of land pollution	Main causes of water pollution
Mobile: cars, buses, planes, trucks, trains **Stationary:** power plants, oil refineries, industrial facilities and factories	➥ Deforestation and consequent erosion ➥ Agriculture, industry, mining, land fills and illegal dumping of waste, as urbanization and construction	➥ Industrial waste ➥ Sewage and waste water ➥ Mining activities, marine dumping ➥ Accidental oil leakage ➥ Burning of fossil fuels

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➥ **Area:** agricultural areas, cities and wood buring fire places ➥ **Natural** - wiid fire wind blow dust, volcanoes.		➥ Chemical fertilizers and pesticides ➥ Leakage form sewer lines.

Table 11.4 Impacts of Air, water and Land Pollution.

Air pollution	Water pollution	Land pollution
➥ Respiratory and health problems ➥ Global warming ➥ Acid rain ➥ Eutrophication ➥ Wildlife destruction ➥ Depletion of the ozone laye	➥ Depletion of drinking water supplies ➥ Agricultural and ecomomic impacts ➥ Food chain distruption ➥ Eutrophication	➥ Health problems ➥ Soil pollution ➥ Water and air pollution ➥ Environmental degradation ➥ Affect tourism industry ➥ Affect wild life

Illustrative Examples

31. One of the following methods is used to reduce air - pollution which one is it?
    A. Recycling of agricultural wastes
    B. Recycling of non - biodegradable materials
    C. Increasing the use of organic fertilizers
    D. Reducing carbondioxide emission

Answer: D

32. What is the direct effect of nitrates and phosphates as water pollutant? They
    A. Cause the death of water organisms
    B. In crease the amount insecticides or herbicides
    C. Accelerate the growth of surface water plants in a lake or river
    D. Allow the spread of water born diseases such as cholera or typhoid.

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Solution

Nitrates and phosphates are valuable nutrients for plants and their discharge into the water systems accelerate the growth of surface - water plants such as algae.

Answer: C

3. Which of the following industrial activities has a minimum potential in polluting the environment?
   A. Making use of petroleum fuel in running industries
   B. Applying phosphate and nitrate fertilizers to maximize crop production
   C. Applying insecticides and herbicides in farming
   D. Making use of hydroelectric power to generate electricity.

Answer: D

4. Which of the following methods is not used to reduce water pollution?
   A. Recycling agricultural and industrial wastes
   B. Discharging heat added water into water systems
   C. Treatment of water before discharge in to water and lakes
   D. Using moderate amount of agricultural chemicals.

Answer: B

5. Among the following pollutants, which pair causes photochemical smog?
   A. Particulates and lead compound
   B. Hydrocarbons and ozone
   C. Nitrogen and sulphur oxides
   D. Carbonmonxide and ozone.

Solution

- Particulates and lead compounds affect breathing system
- Nitrogen and sulphur oxides cause acid rain.

Answer: B

36. The substance that deplete the ozone layer are _____

Solution

Chlorofluorocarbons (CFCs) deplete the ozone layer as follows:

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$$CF_2Cl_2 \longrightarrow CF_2Cl\bullet + Cl\bullet$$

$$Cl\bullet + O_3 \longrightarrow ClO\bullet + O_2$$

37. High levels of _____ cause damage to the brain, kidneys and liver.

Answer: Lead

38. Explain how garbage, plastic and mercury pollute land .

Solution

Garbage - released from homes gets dumped onto the lands and fills it.

Plastics - have the greatest impacts to land

Mercury - disposal of mercury cells pollute the land since mercury and its compounds are toxic.

39. Write the common acid rains

Solution

H_2SO_3 = SO_2 + rain water

H_2SO_4 = SO_3 + rain water

HNO_3 = NO_2 + rain water

40. Define pollution

Solution

Pollution is any discharge of solid, liquid or gaseous substance or radiation /energy/ in to water, air or land that causes unwanted changes.

41. Write some examples of contaminants.

Solution

Water, human, trees and fish bodies

42. The medium that interacts and retains pollutants or converts them chemically is _____

Answer: Sink

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11.4 Global Warming and Climate Change

Green House Gases

A green house gas is a gas that absorbs heat energy and stops heat escaping into space.

Carbon dioxide, methane, nitrous oxide and chlorofluorocarbon are the main green gases

Sources of Green House Gases

1. **Carbondioxide** - is the main green house gas
   - ➥ Its amount has been increasing steadily due to burning fossil fuels in power stations and cars. Smaller amounts of carbondioxide are produced by industrial processes.
2. **Methane** is found in air in smaller amount than carbondioxide. However, it absorbs much more heat energy.
   - ➥ The growing number of cows, pigs and sheep release a lot of methane by bacterial action in their digestive systems.
   - ➥ The decomposition of vegetation, specially in swampy areas, also produces methane as do rice paddy fields. Termites (ant - like insects)also produce methane.
3. **Nitrous oxide** is produced by bacteria in the soil and remains in the atmosphere for a long time
4. **Chlorofluorocarbons (CFCs)** are still used for refrigeration and spray cans although many countries have banned the use of these compounds. Their main effect is not as a green house gas but in the depletion of the ozone layer that surrounds the earth

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The Green House Effect and Global Warming

☑ **Greenhouse gases** prevent earth from cooling down too rapidly when we are not exposed to the sun's rays. If we do not have green house gases in the atmosphere, the earth would be extremely cold.

How Does the Green House Effect Work?

- ➥ Ultraviolte rays have short wavelength and get the earth's atmosphere easily and not absorbed by CO_2.
- ➥ The UV - rays hit the earth's surface and heats up it.

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- Heat energy is lost from the earth's surface as infrared rays (long wavelength radiation).
- Infrared rays can be absorbed by CO_2.
- Some of the heat is re-radiated back to the earth and some escapes in to space.
- The more the amount of CO_2, the more heat is abosorbed. This causes **global warming.**

Climate Change and Its Results

A warmer atmosphere can affect our climate

- The temperature of the air rises. This will melt the polar ice - caps, increases sea levels and lastly causes flooding low - lying areas.
- This may cause less rainfal in some areas results drought.
- There may be more storms and stronger winds that destroys properties and crops.

Illustrative Examples

43. All of the following are greenhouse gases. EXCEPT

A. Carbondioxide
B. Methane
C. Nitrous oxide
D. Carbonmonxiode.

Solution

Green house gases include carbondioxide, nitrous oxide, methane, water vapour and ozone.

Answer: D

44. The decomposition of vegetation in swampy areas produces.

A. Methane
B. carbondioxide
C. Nitrous oxide
D. Water vapour

Answer: A

45. As sulphur dioxide is linked to acid rain, carbondioxide and methane are linked to _____.

Answer: Global Warming

46. What are the effect of global warming?

Solution

An immediate consequence of global warming is the increase in temperature around the world. This melts ice - caps in polar areas and causes flooding, drought and stronger winds.

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47. How green house gases cause global warming?

Solution

By absorbing heat energy radiated from the earth's surface.

48. A gas in the atmosphere that absorbs heat energy is _____ .

Answer: Green house gas.

11.5 Green Chemistry and Cleaner Production

The Twelve principles of Green Chemistry are

1. Prevention

- It is better to prevent waste than to treat or clean up waste after it has been created

2. Atom Economy

- Atom economy = $\frac{\text{mass of desired product}}{\text{total mass of product}} \times 100$
- Synthetic methods should be designed to maximize the incorporation of all materials used in the process in the final product.

3. Less Hazardous Chemical Synthesis

- Whenever practicable synthetic methods should be designed to use and generate substances that possess little or no toxicity to people or the environmet.

4. Designing Safer Chemicals

- Chemical products should be designed to effect their desired function while minimizing their toxicity.

5. Safer Solvents and Auxiliaries

- The use of auxiliary substances (e.g solvents or separation agents) should be made unnecessary whenever possible and innocuous when used.

6. Design for Energy Efficiency

- Energy requirement of chemical processes should be recognized for their environmental and economic impacts and should be

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minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.

7. Use of Renewable Feed Stocks

- A raw material or feed stock should be renewable rather than depleting whenever technically and economically practicable.

8. Reduce Derivatives

- Unnecessary derivatization (use of blocking groups, protection/de-protection and temporary modification/ chemical processes) should be mimized or avoided if possible, because such steps require additional reagents and can generate waste.

9. Catalysis

- Catalytic recgents (as selective as possible) are superior to stoichiometric reagents

10. Design for degradation

- Chemical productions should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.

11. Real - time Analysis for Pollution Prevention

- Analytical methods need to be further developed to allow for real time, in - process monitoring and control prior to the formation of hazardous substances.

12. Inherently Safer Chemistry for Accident Prevention

- Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions and fires.

Green chemistry (Sometimes referred to as sustainable chemistry) is the branch of chemistry that deals with the design and optimization of processes and products in order to lower, or remove altogether, the production and use of toxic substances.

Objectives of Green Chemistry are to:

- Prevent pollution at the molecular level
- Apply to all areas of chemistry
- Apply innovative scientific solutions to real - world environmental problems.

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➥ Result in source reduction because it prevents the generation of pollution.

Illustrative Examples

49. What is atom economy?

Solution

Atom economy is a primary criterion for green chemistry that improve chemical processes, by avoiding the waste of atoms from reactants to produces.

50. Given the reaction: $X + Y \rightarrow W + Z$, W is desired product and Z is by - product. The masses of W and Z formed are 50g and 30g respectively. Calculate the atom economy

Solution

$$\text{Atom economy} = \frac{\text{mass of desired product}}{\text{total mass of products}} \times 100$$

$$= \frac{50g}{(50 + 30)g} \times 100 = 62.5\%$$

51. Write three benefits of green chemistry

Solution

➥ Cleaner production technologies and reduced emissions

➥ Fewer raw materials and natural resources.

➥ Smaller quantities of hazardous waste to be treated and disposed off.

Solved Problems

52. What are the major components of the earth's atmosphere?

Solution

Nitrogen (78%), oxygen (21%) and 1% other gases.

53. What is the gas that supports burning?

Solution

Oxygen

54. _____ is produced as a result of digestion in cows and sheep and from rice paddy fields.

Solution: Methane

55. What air pollutant combines with the haemoglobin of the blood to form

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a very stable compound and so stops the blood transporting oxygen around the body?

Solution: Carbonmonoxide

56. _____ is modification of the environment caused by human influence.

Answer: Pollution

57. Its presence in the stratosphere acts a screen against ultraviolet ray

Solution

It is ozone or trioxygen.

58. Name an air pollutant produced by:

a. Burning of coal

b. Combustion of petrol in a car engine.

Solution

a. Carbondioxide (CO_2)

b. Nitric oxide (NO)

59. What are the four harmful air pollutants produced by motor vehicles?

Solution

- Carbonmonoxide
- Hydrocarbons
- Sulphur dioxide
- Oxides of nitrogen

60. List the common air pollutants

Solution

- Sulphurdioxide
- Hydrocarbons and ozone
- Oxides of nitrogen
- Chlorofluoro carbons
- Carbon monoxide
- Volatile organic compounds
- Particulates
- Polychlorinated bi phenyls.

61. Earth is unique among the planets of our solar system. Why?

Solution

Because it has an atmosphere that is chemically active and rich in oxygen.

62. Why the oxygen cycle is complicated? Describe it by using diagram.

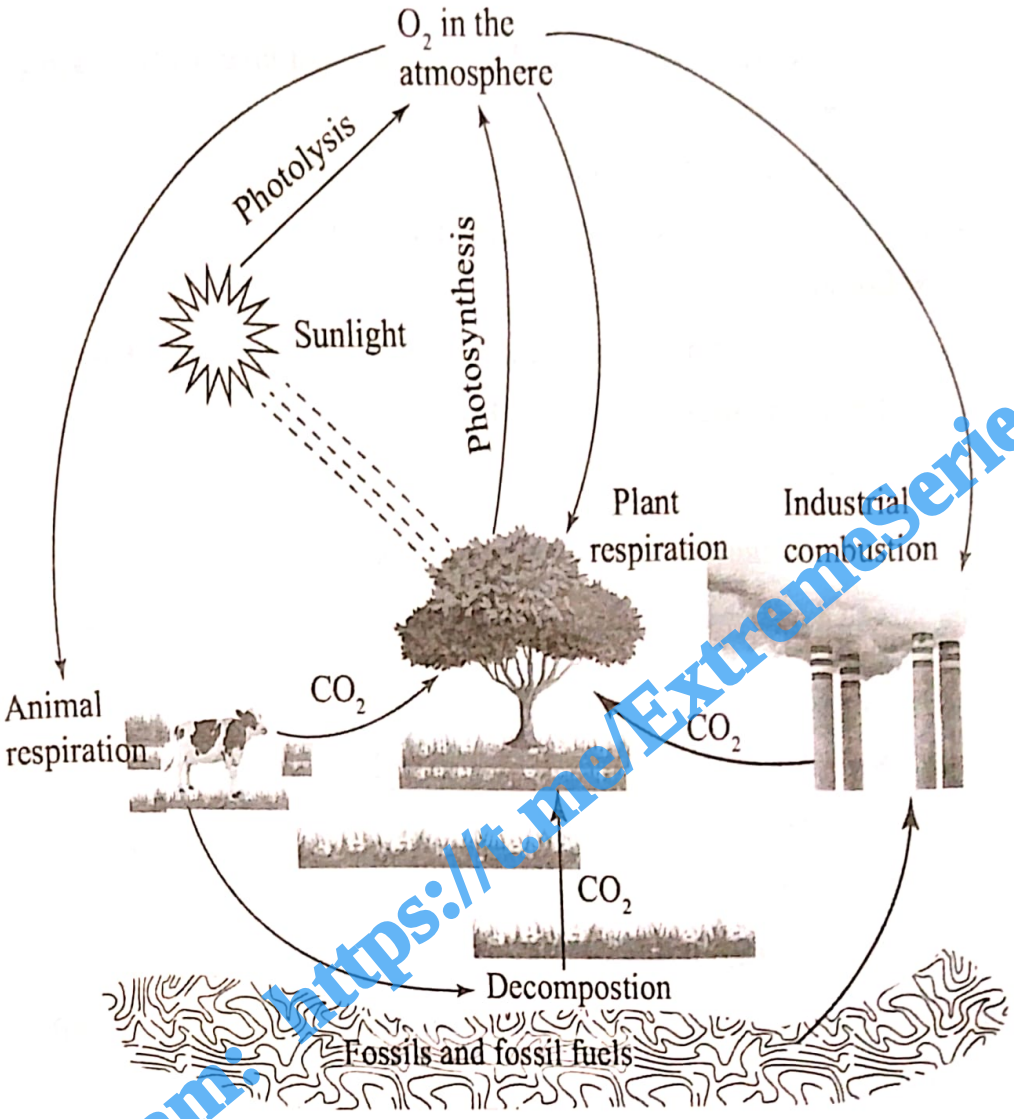
Solution

The cycle is complicated because oxygen appears in so many chemical forms and combinations, primarly as molecule oxygen, in

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water and in organic and inorganic compounds.



63. Describe the four layers of the atmosphere

Solution

Troposphere - Closest to earth's surface

- ➥ Temperature decreases with increasing altitude
- ➥ Contains 80% of gases and all water vapour
- ➥ Thinnest layer of the atmosphere (10km)
- ➥ Rain, lightning, hurricanes occur.

Stratosphere - Contains nitrogen oxygen and ozone

- ➥ Temperature increases with altitude
- ➥ 10km - 50km

Mesosphere - low ozone and other gases concentration

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- Temperature decreases with increasing altitude.
- 50km - 80km

Thermosphere or ionosphere - uper most layer of the atmosphere

- 80km - 500km
- Temperature increases with altitude

64. List some processes that yield carbondioxide

Solution

1. Complete combustion of carbon containing substance with oxygen.
2. Decomposition of many carbonates

$$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$$

3. The reaction between acids and carbonates

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

4. By - product of the fermentation process

$$C_6H_{12}O_6\ (aq) \xrightarrow{yeast} 2C_2H_5OH(aq) + 2CO_2(g)$$

glucose

5. Respiration

$$C_6H_{12}O_6\ (aq) \longrightarrow 6O_2(g) + 6H_2O(l)$$

65. How does an increase world population enhance the greenhouse effect?

Solution

An increase in world population increases the concentration of green house gases in the atmosphere The main causes of are:

- Burning fossil fuels for heat and electricily
- Fossil fuel use in transport
- Agriculture
- Deforestation
- Cement manufacture

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Supplementary Exercise

I. Multiple choice items

1. Which of the following can act as a greenhouse gas?
   A. O_2 B. N_2 C. CO_2 D. Ar
2. The greenhouse of stratospheric ozone
   A. Caused by depletion of stratospheric ozone
   B. Entirely the result of human activity
   C. A nature phenomenon that has been enhanced by human activity
   D. The absorption of the sun's energy molecules and atoms in the upper atmosphere.
3. All of the following are impacts of acid rain **EXCEPT**
   A. Death of fish and other aquatic species
   B. Corrosion of water pipes
   C. Respiratory issues in animals and humans
   D. Increasing the temperature of an environment
4. Photochemical smog is related to pollution of
   A. Air B. Water C. Land D. None
5. Most hazardous metal pollutant of automobile exhaust is
   A. Hg B. Pb C. Sn D. Pd
6. Environmental pollution affects.
   A. Biotic components C. Animals only
   B. Plants only D. Both biotic and abiotic components
7. Which of the following is major sink for carbon monoxide?
   A. Water C. Animals respiration
   B. Soil D. Salts
8. Eutrophication is process which involves
   A. Depletion of ozone layer
   B. Increase in the amount of ozone
   C. Decrease in amount of O_2 in H_2O
   D. Decreases in amount SO_2
9. Which one of the following is a source of energy but does not cause

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pollution?

A. Gasoline
C. Fossil fuels
B. Nuclear power plant
D. Sun

10. Which of the following is not a chemical pollutant?

A. Solid waste
C. Noise
B. Insecticides
D. Liquid waste

11. Earth is protected from U.V radiations by

A. CO_2 layer
C. Troposphere
B. O_2 layer
D. O_3 layer

12. Water pollution is mainly due to which of the following?

A. Sulphurdioxide
C. Industrial effluents
B. Oxygen
D. Carbodioxide

13. Green chemistry refers to

A. Chemistry of plants
B. Development of chemical products and processes less harmful to humans
C. Chemistry of green pigments
D. Chemistry of green house effect

14. Addition of phosphate and nitrate fertilizers in to water leads to

A. Increased growth of decomposers
B. Reduced growth of algae
C. Increased growth of algae
D. Nutrient enrichment

15. Air pollution is not caused by

A. Pollen grains
C. Industries
B. Hydroelectric power
D. Automobiles

II. Short answer items.

16. Any substance produced either by natural source or by human activity which causes adverse effect on environment is _____

17. List some soil pollutants.

18. The purpose of nitrogen is _____

19. The most abundant noble gas in the atmosphere is _____

20. Mention three diseases that are caused by drinking water

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Answer to supplementary exercise.

Part II

1. C	3. D	5. B	7. B	9. D	11.D	13.B	15.B
2. B	4. A	6. D	8. C	10.C	12.C	14.C	

Pare II

16. Pollutant
17. Insecticides, fungicides, herbicides
18. To make protein
19. Argon
20. Cholera, typhoid, dysentery

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Absolute temperature scale: A temperature scale that uses the absolute zero of temperature as the lowest temperature.

Absolute zero: Theoretically the lowest attainable temperature.

Accuracy: is the agreement between the measured quantity and the accepted value.

Acid – base reaction: A reaction between an acid and a base.

Acid: is a compound that ionizes in water to give hydronium ion and anion.

Activation energy: is the energy that must be absorbed by a system to cause it to react.

Aldehydes: Compounds with a carbonyl functional group and the general formula RCHO, where R is an H atom, an alkyl, or an aromatic group.

Aliphatic hydrocarbons: Hydrocarbons that do not contain the benzene group or the benzene ring.

Alkali metals: The Group 1A elements (Li, Na, K, Rb, Cs, and Fr).

Alkaline earth metals: The group 2A elements (Be, Mg, Ca, Sr, Ba and Ra).

Alpha particle: is positively charged particle ejected at high speeds from certain radioactive substances.

Amorphous solid: is solid without long range order.

Amphiprotic substance: A substance that can behave as either a Bronsted acid or a Bronsted base.

Amplitude: is the maximum height of a wave as measured from the axis of propagation.

Anode: is electrode at which oxidation occurs.

Antibonding molecular orbital: A molecular orbital in which the energy of the electrons is higher than that of all the parent orbital electrons.

Aromatic hydrocarbon: A hydrocarbon that contains one or more benzene rings.

Atom: is the smallest particle of an element that retains the properties of that element.

Atomic mass unit (amu): The unit of a scale of relative atomic masses of the elements.

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Atomic orbital: the wave for an allowed energy state of an electron is an atom.

Atomic theory: is a theory that describes the structure and behavior of substances.

Aufbau principle: As protons are added one by one to the nucleus to build up the elements, electrons similarly are added to the atomic orbitals.

Band theory: Delocalized electrons move freely through "bands" formed by overlapping molecular orbitals.

Base: is a compound that ionizes in water to give hydroxide ion and a cation.

Battery: is an electrochemical device that produces a current or flow of electrons at a constant voltage.

Beta: particle electrons ejected at high speeds from certain radioactive nuclei.

Boiling point: the temperature at which the vapor pressure of a liquid is equal to the external pressure on the liquid.

Bond energy: the energy that must be supplied to a gaseous molecule to separate two of its atoms.

Bond length: The distance between the centers of two bonded atoms in a molecule.

Bond order: The number of bonding electron pairs shared by two atoms in a molecule.

Bond: angle is the angle between two atoms bonded to a central atom in a molecule.

Bonding molecular orbital: A molecular orbital in which the energy of the electrons is lower than that of the parent orbital electrons.

Bronsted acid: A substance capable of donating a proton.

Bronsted base: A substance capable of accepting a proton.

Buffer solution: A solution of (a) a weak acid or base and (b) its salt; both components must be present. The solution has the ability to resist changes in pH upon the addition of small amounts of either acid or base.

Buffer: A solution that resists change in pH when hydroxide or hydronium

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ions are added.

Carbohydrates: A polyhydroxyaldehyde or polyhydroxy ketone.

Carbonyl group: the functional group characterizing aldehydes and ketones.

Carboxyl group: the functional group characterizing organic acids.

Catalyst: a reagent that can increase the rate of a reaction while not being consumed in the reaction.

Cathode: the electrode of an electro chemical cell at which reduction occurs.

Cell potential (Eecll): A measure in volts of the tendency of reactant to proceed to products in an electro chemical cell.

Chemical equilibrium: At state in which the rates of the forward and reverse reactions are equal.

Chemical kinetics: is the study of the rate of chemical reactions under various conditions and of reaction mechanisms.

Chemical kinetics: The area of chemistry with the speeds, or rates, at which chemical reactions occur.

Chlo – alkali process: The production of chlorine gas by the electrolysis of aqueous NaCl solution.

Colligative properties: properties of solution that depend only on the number of solute particle per solvent molecule and not on the nature of the solute or solvent.

Collision theory: is a theory of reaction rates that assumes molecules must collide in order to react.

Colloid: a state intermediate between a solution and a suspension in which solute particles are large enough to scatter light but too sm ettle out.

Common: ion effect the limitation on acid or base dissociation caused by addition of some species common to the equilibrium.

Conjugate acid: base pair A pair of molecules or ions related by gain or loss of a hydrogen ion.

Coordinate covalent bond: A bond in which the pair of electrons is supplied by one of the two bonded atoms; also called a dative bond.

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Corrosion: is a deterioration of metals by an oxidation – reduction reaction.

Diamagnetism: the physical property of being repelled by a magnetic field.

Diffusion: the gradual mixing of the molecules of two or more substances owing to random molecular motion.

Dipole – dipole forces: Forces that act between polar molecules.

Diprotic acid: Each unit of the acid yields two hydrogen ions upon ionization.

Dispersion forces: The attractive forces that arise as a result of temporary dipoles induced in the atoms or molecules; also called **London forces.**

Double bond: a bond formed by sharing two pairs of electrons, one pair in a sigma and the other in a pi bonds.

Dynamic equilibrium: is chemical equilibrium that stresses the on going nature of the processes involved.

Electrode: is a device such as a metal plate or wire for conducting electrons into and out of solutions.

Electrolysis: the use of electrical energy to produce chemical change.

Electrolyte: A substance that ionizes in water or on melting to form an electrically conducting solution.

Electron density: The probability that an electron will be found at a particular region in an atomic orbital .

Electron: a negatively charged subatomic particle found in space about the nucleus.

Endothermic process: a thermodynamics process in which heat flows into a system from its surroundings.

Energy: the capacity to do work and transfer of heat.

Enthalpy change (ΔH): heat transferred at constant pressure.

Equilibrium: A condition in which the forward and reverse reaction rates in a physical or chemical system are equal.

Equivalence point: the point in a titration at which one reactant has been

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exactly consumed by addition of the other constant.

Excited state: A stationary state in which the energy is higher than in the ground state.

Exothermic process: A thermodynamic process in which heat flows from a system to its surroundings.

Extensive properties: physical properties that depend on the amount of matter present.

Fat: a solid triester of an acid with glycerol.

Ferromagnetism: a form of paramagnetism seen in some metals and their alloys in which the magnetic effect is greatly enhanced.

First – order reaction: A reaction whose rate depends on reactantconcentration raised to the first power.

Formal charge: The differentiable between the valence electrons in an isolated atom and the number of electrons assigned to that atom in a Lewis structure.

Frequency: the number of complete waves passing a point in a given amount of time.

Fuel cell: a voltaic cell in which reactants are continuously added or which can be restored after discharge by the addition of reactants.

Gamma rays: the most penetrating form of radiation, having higherenergies than those of x – rays.

Gibbs free energy: G a thermodynamics state function that relates enthalpy, temperature and entropy.

Greenhouse effect: Carbon dioxide and other gases' influence on Earth's temperature.

Ground state: the lowest energy stationary state.

Half – cell reaction: Oxidation and reduction reactions at the electrodes.

Half – life: The time required for the concentration of one of the reactant to reach half of its initial value.

Half – reactions: the two chemical equations into which the equation for divided one representing the oxidation process and the other the reeducation process.

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Hess's law: When reactants are converted to products, the change in enthalpy is the same whether the reaction takes place in one step or in a series of steps.

Heterogeneous equilibrium: An equilibrium state in which the reacting species are not all in the same phase.

Heterogeneous mixture: The individual components of a mixture remain physically separated and can be seen as separate components.

Hund's rule: The most stable arrangement of electrons in subshells is the one with the greatest number of parallel spins.

Hybrid Orbitals: Atomic orbitals obtained when two or more nonequivalent orbitals of the same atom combine.

Hybridization: The process of mixing the atomic orbitals in an atom (usually the central atom) to generate a set of new atomic orbitals.

Hydrogen bond: A special type of dipole – dipole interaction between the hydrogen atom bonded to an atom of a very electronegative element (F, N, O) and another atom of one of the three electronegative elements.

Hydrolysis: A reaction between an ion and water in which an oxygen – hydrogen bond is broken.

Hypothesis: a tentative explanation of or production derived from experimental observations.

Ideal solution: is a solution that obeys Raoult's law.

Intensive properties: physical properties that do not depend on theamount of matter present.

Intermoeluclar forces: interactions between molecules, between ions or between molecules and ions.

Ion – dipole force: the attraction between an ion and a polar molecule.

Ion: an atom or group of atoms that has lost or gained one or more electrons.

Ionization energy: The minimum energy required to remove an electron from an isolated atom (or an ion) in its ground state.

Kinetic energy: the energy of a moving object dependent on its mass and velocity.

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Lattice energy: The energy required to completely separate one mole of a solid ionic compound into gaseous ions.

Law: a concise verbal or mathematical statement of a relation that is always the same under the same conditions.

Leaching: a method of separating a desired element from its ore by forming a soluble compound.

Lewis acid: A substance that can accept a pair of electrons.

Lewis base: A substance that can donate a pair of electrons.

Ligand: A molecular or an ion that is bonded to the metal ion in a complex ion.

Line spectrum: a spectrum obtained when light is separated into its various wavelengths each of which is defected as a separate line.

London forces: very weak intermolecular forces occurring between two induced dipoles.

Melting point: the temperature at which the crystal lattice structure of a solid collapses and solid is converted to liquid.

Mixture: A contribution of two or more substances in which each substance retains its identity.

Molality (m): the number of moles of solute per kilogram of solvent.

Molar solubility: The number of moles of solute in one liter of a saturated solution (mol/L).

Molarity (M): The number of moles of solute per liter of solution.

Mole (mol): The amount of substance that contains as many elementary entities (atoms, molecules, or other particles) as there are atoms in exactly 12 grams (or 0.012 kilograms) of the carbon - 12 isotope.

Mole fraction (X): The ratio of the moles of a certain substance to the total moles in a mixture of substances.

Molecular geometry: the three – dimentional arrangement of atoms in a molecule.

Molecular orbital: An orbital that results from the interaction of the atomic orbitals of the bonding atoms.

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Moleculatiry: the number of particles colliding in an elementary process.

Monomer: is the small unit from which a polymer is constructed.

Nernst equation: A mathematical expression that relates the potential of an electro chemical cell to the concentration of the cell reactants or products.

Net ionic equation: A chemical equation in which only those ions undergoing chemical changes in the course of the reaction are represented.

Neutralization: An acid – base reaction that produces a neutral solution of salt and water.

Neutron: An electrically neutral subatomic particle found in the nucleus.

Non electrolyte: a substance dissolves in water to form an electrically non – conducting solution.

Non polar molecules: A molecule that does not possess a dipole moment.

Nucleus: the core of an atom made up of protons and neutrons.

Octet Rule: An atom other than hydrogen tends to form bonds until it is surrounded by eight valance electron.

Open System: A system that can exchange mass and energy (usually in the form of heat) with its surroundings.

Orbital hybridization: the combination of atomic orbitals to produce equivalent hybrid orbitals that are specialized for bonding.

Orbital: the wave for an allowed energy state of an electron in atom or molecule.

Ore a substance: containing a desired mineral or element usually with large quantities of impurities such as sand and clay.

Osmosis: the movement of solvent molecules through a semi permeable membrane from a region of lower solute concentration to a region of higher solute concentration.

Osomotic pressure (π): the force exerted by solvent molecules passing through a semi permeable in a solution system at equilibrium.

Oxidation: the loss of electrons by an atom, ion or molecule leading to an increase in oxidation number.

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Paramagnetism: the physical property of being attracted by a magnetic field.

Peptide bond: the amide linkage between two amino acid molecules.

Phase diagram: a graph showing what poles of a substance exist at various temperatures and pressures.

Photoelectric effect: the ejection of electrons by a metal bombarded with light of at least a critical minimum frequency.

Pi bond (π): A covalent bond formed by sideways overlapping orbitals; its electron density is concentrated above and below the plane of the nuclei of the bonding atoms.

Pi molecular orbitals: A molecular orbital in which the electron density is concentrated above and below the line joining the two nuclei of the bonding atoms.

Polar covalent bond: In such a bond, the electrons spend more time in the vicinity of one atom than the other.

Polar molecule: A molecule for measuring the rotation of polarized light by optical isomers.

Polymer: a large molecule composed of many smaller repeating units.

Positron: A nuclear particle having the same mass as an electron but a positive charge.

Precision: the agreement of repeated measurements of a quantity with one another.

Proton: positively charged subatomic particle found in the nucleus.

Quantum numbers: a set of numbers with integer values that define the properties of an atomic orbital.

Quantum: the minimum energy change in an atom emitting or absorbing radiation.

Radioactivity: the spontaneous emission of energy without the absorption of energy.

Rate Law: An expression relating the rate of a reaction to the rate constant and the concentrations of the reactants.

Reaction intermediate: a species produced in one step of a reaction

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sequence and then consumed in a subsequent step.

Reaction mechanisms: the sequence of events at the molecular level that control the speed and outcome of a reaction.

Reaction Order: The sum of the powers to which all reactantn concentrations appearing in the rate law are raised.

Reaction rate: The change in the concentration of reactant or product with time.

Reduction: The gain of electrons by an atom, ion or molecule leading to a decrease in its oxidation state.

Resonance structure: The possible structures of a molecule for which more than one Lewis structure can be written, differing by the number of bond pairs between a given pair of atoms.

Reversible reaction: A reaction that can occur in both directions.

Salt bridge: a device for maintaining a balance of ion charges in the compartments of an electrochemical cell.

Salt: an ionic compound whose cation comes from a base and whose an ion comes from an acid.

Saturated solution: a stable solution in which the maximum amount of solute has been dissolved.

Second - order reaction: A reaction whose rate depends on reactant concentration raised to the second power or on the concentrations of two different reactants, each raised to the first power.

Sigma bond (σ): A covalent bond formed by orbitals overlapping end - to - end; its electron density is concentrated between the nuclei of the bonding atoms.

Sigma molecular orbital: A molecular orbital in which the electron density is concentrated around a line between the two nuclei of the bonding atoms.

Significant figures: The number of meaningful digits in a measured or calculated quantity.

Single bond: Two atoms are held together by one electron pair.

Solute: the material dissolved in a solution.

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Solution: a homogeneous mixture of two or more substances in a single phase.

Solvent: the medium in which a solute is dissolved to form a solution.

Standard enthalpy of formation ($\Delta H°f$): The heat change that results when one mole of a compound is formed from its elements in their standard states.

Standard entropy of reaction ($\Delta S°rxn$): The entropy change when the reaction is carried out under standard - state conditions.

Standard free - energy of reaction ($\Delta G°f$): The free - energy change when 1 mole fo a compound is synthesized from its elements in their standard states.

Stationary state: any of the allowed energy states of the single electron in a hydrogen atom.

Temperature: a physical property that determine the direction of heat flow in an object on contact with anther object.

Theory: a unifying principle that explains a body of facts and the law based on them.

Thermodynamics: the science of heat or energy flow in chemical reactions.

Titration: a procedure for the quantitative analysis of a substance by means of an essentially complete reaction in solution with a reagent of known concentration.

Triple bond: a bond formed by sharing three pairs of electrons, one is a sigma and the other two in pi bonds.

Triple point: the temperature and pressure at which the solid, liquid and vapor phases of a substance are in equilibrium.

Universe: the sum of the system and its surroundings.

Unsaturated solution: a solution in which the concentration of dissolved solute is less than the saturation amount.

Valence bond theory: a view of bonding in which bonds arise from the overlap of atomic orbitals on two atoms to give a bonding orbital with electrons localized between the bonded atoms.

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Valence shell electron: pair repulsion (VSEPR) theory a model of structure about an atom in a molecule in which structural electron pairs are arranged so as to maximize the angles between them.

Van der Walls forces: weak intermolecular forces not involving ions.

Volatility: the tendency of the molecules of a substance to escape from the liquid phase and enter the gas phase.

Wave length: the distance between successive crests or troughs in a wave.

Weak acids: Acids that ionize only to a limited extent in water.

Weak bases: Bases that ionize only to a limited extent in water.

Work: they product of force time distance over which the force acts.

Zwitter ion: a dipolar ion.

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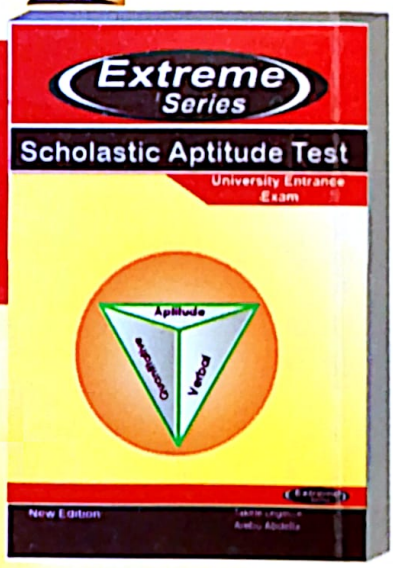
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